Practice 1

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Create a database & use:

```
1     CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 •   USE bookflow_db;
```

Output

Action Output

#	Time	Action
1	09:08:46	SHOW DATABASES
2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
3	09:09:40	USE bookflow_db

Create a physician table:

```
3 •  CREATE TABLE physician (
4        employeeid INT PRIMARY KEY,
5        name VARCHAR(100),
6        position VARCHAR(100),
7        ssn VARCHAR(20)
8    );
```

Output

Action Output

#	Time	Action	Mes
1	14:56:17	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 rov
2	14:56:29	USE bookflow5_db	0 rov
3	14:59:46	CREATE TABLE physician (employeeid INT PRIMARY KEY, name V...	0 rov

Insert data into physician:

```
9  INSERT INTO physician VALUES
10 (1,'Likhitha','Staff Internist','111111111'),
11 (2,'Kavya','Attending Physician','222222222'),
12 (3,'Divya','Surgical Attending Physician','333333333'),
13 (4,'Akhila','Senior Attending Physician','444444444'),
14 (5,'Sushma','Head Chief of Medicine','555555555'),
15 (6,'Vaishnavi','Surgical Attending Physician','666666666'),
16 (7,'Navya','Surgical Attending Physician','777777777'),
17 (8,'Sai','MD Resident','888888888'),
18 (9,'Shiva','Attending Psychiatrist','999999999');
19
20
```

Output

Action Output

#	Time	Action	Message
14	15:03:56	DROP DATABASE bookflow5_db	1 row(s) affected
15	15:04:01	DROP DATABASE bookflow6_db	0 row(s) affected
16	15:04:07	CREATE DATABASE IF NOT EXISTS bookflow6_db	1 row(s) affected
17	15:04:09	USE bookflow5_db	Error Code: 1049. Ur
18	15:04:14	USE bookflow6_db	0 row(s) affected
19	15:04:17	CREATE TABLE physician (employeeid INT PRIMARY KEY, nam...	0 row(s) affected
20	15:04:19	INSERT INTO physician VALUES (1,'Likhitha','Staff Internist','11111111...	9 row(s) affected Re

Select:

```
19 SELECT * FROM physician;
20
21
```

Result Grid | Filter Rows: | Edit: | Expor

employeeid	name	position	ssn
1	Likhitha	Staff Internist	111111111
2	Kavya	Attending Physician	222222222
3	Divya	Surgical Attending Physician	333333333
4	Akhila	Senior Attending Physician	444444444
5	Sushma	Head Chief of Medicine	555555555
6	Vaishnavi	Surgical Attending Physician	666666666
7	Navya	Surgical Attending Physician	777777777
8	Sai	MD Resident	888888888
9	Shiva	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Create department table:

```sql
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

Output

Action Output

#	Time	Action
20	15:04:19	INSERT INTO physician VALUES (1,'Likhitha','Staff Internist','11111111...
21	15:04:57	SELECT * FROM physician LIMIT 0, 1000
22	15:05:39	CREATE TABLE department (departmentid INT PRIMARY KEY, n...

Insert data into department:

```sql
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
SELECT * FROM department;
```

Result Grid | Filter Rows:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

Inner join

```sql
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

Result Grid | Filter Rows:

Department	Head_Physician
General Medicine	Akhila
Surgery	Navya
Psychiatry	Shiva

Where clause:

```sql
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
```

Result Grid | Filter Rows: | Edit:

employeeid	name	position	ssn
3	Divya	Surgical Attending Physician	333333333
6	Vaishnavi	Surgical Attending Physician	666666666
7	Navya	Surgical Attending Physician	777777777

Where + Like:

```
38 •  SELECT *
39    FROM physician
40    WHERE name LIKE 'Likhitha%';
```

Result Grid | Filter Rows: | Ed

employeeid	name	position	ssn
1	Likhitha	Staff Internist	111111111
NULL	NULL	NULL	NULL

Count():

```
41 •  SELECT COUNT(*) AS Total_Physicians
42    FROM physician;
```

Result Grid | Filter Rows: | Expo

Total_Physicians
9

Count Distinct:

```
43 •  SELECT COUNT(DISTINCT position)
44    FROM physician;
```

Result Grid | Filter Rows:

COUNT(DISTINCT position)
7

Group by:

```
45 •  SELECT position,
46           COUNT(*) AS Total_Employees
47    FROM physician
48    GROUP BY position;
```

Result Grid | Filter Rows: | Exp

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Group by + having:

```
49 •  SELECT position,
50            COUNT(*) AS Total_Employees
51    FROM physician
52    GROUP BY position
53    HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Expor

position	Total_Employees
Surgical Attending Physician	3

Order by:

```
54 •  SELECT *
55    FROM physician
56    ORDER BY name ASC;
```

Result Grid | Filter Rows: | Edit:

employeeid	name	position	ssn
4	Akhila	Senior Attending Physician	444444444
3	Divya	Surgical Attending Physician	333333333
2	Kavya	Attending Physician	222222222
1	Likhitha	Staff Internist	111111111
7	Navya	Surgical Attending Physician	777777777
8	Sai	MD Resident	888888888
9	Shiva	Attending Psychiatrist	999999999
5	Sushma	Head Chief of Medicine	555555555
6	Vaishnavi	Surgical Attending Physician	666666666
NULL	NULL	NULL	NULL

Order by desc:

```
57 •  SELECT *
58    FROM physician
59    ORDER BY employeeid DESC;
```

Result Grid | Filter Rows: | Edit:

employeeid	name	position	ssn
9	Shiva	Attending Psychiatrist	999999999
8	Sai	MD Resident	888888888
7	Navya	Surgical Attending Physician	777777777
6	Vaishnavi	Surgical Attending Physician	666666666
5	Sushma	Head Chief of Medicine	555555555
4	Akhila	Senior Attending Physician	444444444
3	Divya	Surgical Attending Physician	333333333
2	Kavya	Attending Physician	222222222
1	Likhitha	Staff Internist	111111111
NULL	NULL	NULL	NULL

Where + in:

```
60 •  SELECT *
61    FROM physician
62    WHERE employeeid IN
63    (
64        SELECT head
65        FROM department
66    );
```

Result Grid | Filter Rows: | Edit:

employeeid	name	position	ssn
4	Akhila	Senior Attending Physician	444444444
7	Navya	Surgical Attending Physician	777777777
9	Shiva	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Subquery:

```
67 •  SELECT *
68    FROM physician
69    WHERE employeeid NOT IN
70    (
71        SELECT head
72        FROM department
73    );
```

Result Grid | Filter Rows: | Edit:

employeeid	name	position	ssn
1	Likhitha	Staff Internist	111111111
2	Kavya	Attending Physician	222222222
3	Divya	Surgical Attending Physician	333333333
5	Sushma	Head Chief of Medicine	555555555
6	Vaishnavi	Surgical Attending Physician	666666666
8	Sai	MD Resident	888888888
NULL	NULL	NULL	NULL

Join + where

```
74 •  SELECT p.name
75    FROM physician p
76    JOIN department d
77    ON p.employeeid=d.head
78    WHERE d.name='Surgery';
```

Result Grid | Filter Rows:

name
Navya

Join + order by:

```sql
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

Result Grid | Filter Rows:

name	name
General Medicine	Akhila
Psychiatry	Shiva
Surgery	Navya

Count + having

```sql
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Result Grid | Filter Rows:

position	Total
Surgical Attending Physician	3

Exists:

```sql
SELECT *
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.head
);
```

Result Grid | Filter Rows:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

Max():

```
98 •  SELECT MAX(employeeid)
99    FROM physician;
```

Result Grid | Filter Rows:

	MAX(employeeid)
▶	9

Min():

```
100 •  SELECT MIN(employeeid)
101    FROM physician;
```

Result Grid | Filter Rows:

	MIN(employeeid)
▶	1